

Task 1 — Create a Linux Live USB

Using Rufus on Windows

1. Download the Linux ISO from its official website.
2. Insert an empty USB drive (8 GB+ is a comfortable choice).
3. Open Rufus.
4. Select your USB under Device.
5. Select the downloaded ISO under Boot selection.
6. Keep the recommended partition settings.
7. Click Start.
8. Reboot the computer.
9. Open the boot menu — commonly F12, F9, Esc, or F8, depending on the manufacturer.
10. Select the USB drive.
11. Choose Try Ubuntu / Live rather than installing it.

Task 2 — Access the Windows partition

From the Linux live desktop:

1. Open Files.
2. Look under Other Locations.
3. Select the Windows NTFS partition.
4. Browse to:
 - Users\<Windows-username>\

The exact WhatsApp Desktop location depends on the WhatsApp version and whether it was installed through the Microsoft Store or as a traditional desktop application.

For a Store-style installation, application data can be under locations such as:

- Users\<username>\AppData\Local\Packages\

Look for a WhatsApp-related package rather than assuming one fixed folder name.

Three categories of data you may encounter

Depending on the WhatsApp version, installation method, and encryption/protection:

1. Application/cache data
2. Media-related files or thumbnails
3. Local application databases/configuration data

Task 3 — Offline password attack simulation

Windows stores local account password information in the SAM (Security Account Manager) database, normally located at:

- C:\Windows\System32\config\SAM

However, simply copying a real SAM database from another Windows installation and cracking its hashes can expose actual credentials. Instead, demonstrate the same concept using lab-created password hashes.

Create a practice hash

For example, create an MD5 hash of a deliberately fake password:

- LabPassword123

You can then use a password-cracking tool such as John the Ripper against that test hash.

Example:

- `echo -n "LabPassword123" | md5sum`

You'll get something similar to:

- `<hash> -`

Put the hash into:

- `lab_hash.txt`

Then John can be used against your deliberately generated lab hash:

- `john --format=raw-md5 lab_hash.txt`

And:

- `john --show lab_hash.txt`

Task 4 — Mount and unmount NTFS

First identify the partitions:

- `lsblk`

You may see something like:

```
nvme0n1
├─nvme0n1p1
├─nvme0n1p2
└─nvme0n1p3
```

Find the NTFS partition:

- `lsblk -f`

Suppose your Windows partition is:

- `/dev/nvme0n1p3`

Create a mount point:

- `sudo mkdir /mnt/windows`

Mount it:

- `sudo mount -t ntfs3 /dev/nvme0n1p3 /mnt/windows`

Check:

- `ls /mnt/windows`

Unmount

First leave the directory:

- `cd ~`

Then:

- `sudo umount /mnt/windows`

Verify:

- `lsblk`

Task 5 — Real-world examples

Three relevant categories/cases to research are:

Case	Relevance
------	-----------

Kon-Boot-style attacks	Bootable environments have been used to bypass or alter local authentication in certain circumstances when an attacker has physical access.
Offline Windows password-recovery attacks	Bootable Linux environments have historically been used to access Windows disks and perform offline password auditing/recovery.
Forensic/incident-response abuse	Bootable Linux/WinPE environments can be used to access disks without starting the installed operating system, which can be abused when an attacker has physical access.